Discovering Dynamical Laws with Kolmogorov–Arnold Networks

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The problem and the KAN-ODE starting point

The inverse problem

Observed paths show how a system evolves. We want to infer the law that generates those paths and simulate it from new initial conditions.

Koenig, Kim and Deng (2024): KAN-ODEs

A KAN represents an unknown state-to-rate map. An ODE solver turns it into predicted paths, and training minimizes their mismatch with observations.

The question in this talk

In a synthetic economy, keep the accounting relations and learn two responses. Check function capacity with a separate direct-fit diagnostic, then recover the law from state paths and test it from unseen initial conditions.

Three states describe the economy

State	Meaning
$\omega = \dfrac{wL}{Y}$	Wage share: fraction of output paid as wages.
$e = \dfrac{L}{N}$	Employment rate: employed share of labor force.
$d = \dfrac{D}{Y}$	Debt relative to annual output.

w: real wage; L: employment; N: labor force; D: debt; Y: real output.

$$u(t) = \big(\omega(t), e(t), d(t)\big), \qquad t \text{ measured in model years.}$$

The observed data are paths of these three quantities.

Two unknown responses create the feedback

Functions to learn	Meaning
$\Phi(e) = \dfrac{\dot{w}}{w}$	Wage growth responds to employment.
$\kappa(\pi) = \dfrac{I}{Y}$	Investment responds to profit.

Accounting relations	Meaning
$\pi = 1 - \omega - rd$	Profit share after wages and debt interest.
$g(\pi) = \dfrac{\kappa(\pi)}{\nu} - \delta$	Output growth, $g = \dot{Y}/Y$.

I: investment.

Baseline constants: $r = 0.030$ (interest rate), $\nu = 3$ (capital/output), $\delta = 0.010$ (depreciation rate).

The economic dynamics form a coupled ODE

Grasselli & Costa Lima (2012): *An analysis of the Keen model*.

State equation	**Interpretation**
$\dot{\omega} = \omega[\Phi(e) - \alpha]$	Wage growth relative to productivity.
$\dot{e} = e[g(\pi) - \alpha - \beta]$	Output growth changes labor demand.
$\dot{d} = \kappa(\pi) - \pi - dg(\pi)$	New borrowing minus dilution by growth.

Baseline growth rates: $\alpha = 0.025$ for productivity and $\beta = 0.020$ for the labor force.

Initial-value problem

$$\dot{u} = F(u; \Phi, \kappa), \qquad u(0) = u_0 \in \Omega,$$

$$\Omega = \{(\omega, e, d) : \omega > 0,\ 0 < e < 1,\ d \in \mathbb{R}\}.$$

Picard–Lindelöf: C^1 responses make F locally Lipschitz in Ω, giving a unique local solution.

Simulated data and measurement noise

Sample reference ODE paths, then add measurement noise

$$y_{m,i} = u_m^*(t_i) + \varepsilon_{m,i}, \qquad t_i = i\Delta t, \quad \Delta t = 0.1 \text{ years}.$$

u_m^*: simulated reference path m; $y_{m,i}$: recorded state at observation i; $\varepsilon_{m,i}$: measurement error.

State scales: one standard deviation per variable

$$s_\omega = \mathrm{SD}(\omega), \quad s_e = \mathrm{SD}(e), \quad s_d = \mathrm{SD}(d), \quad S = \mathrm{diag}(s_\omega, s_e, s_d).$$

Each SD is computed over clean training states. The same S normalizes trajectory errors.

$$\varepsilon_{m,i} \sim \mathcal{N}(0, \eta^2 S^2), \qquad \eta \in \{0, 0.01\}.$$

Purpose of η: we choose the noise level to test how reliably the unknown responses can be recovered from imperfect state observations.

Data split

48 training paths $\mathcal{D}_{\text{train}}$, 12 validation paths $\mathcal{D}_{\text{val}}$ and 16 test paths.

Two KANs learn the two economic responses

Both $[1, 4, 1]$ KANs have one input, four hidden nodes and one output; c collects their separate edge coefficients and node biases.

Step	Wage growth $\widehat{\Phi}(e; c)$	Investment $\widehat{\kappa}(\pi; c)$
Prepared input	$z_e = A_e[-\log(1-e)]$	$z_\pi = A_\pi[\pi]$
Apply learned KAN	$h_\Phi(z_e; c)$	$h_\kappa(z_\pi; c)$
Return response	$0.025 + 0.1 \sinh h_\Phi(z_e; c)$	$0.165 + 0.3 \sinh h_\kappa(z_\pi; c)$

A_e, A_π: fixed rescalings to $[-1, 1]$.

Use $(h, z) = (h_\Phi, z_e)$ for wages or $(h, z) = (h_\kappa, z_\pi)$ for investment.

$$z_j^{(1)} = b_j^{(1)} + \psi_{j1}^{(1)}(z), \qquad j = 1, \ldots, 4,$$

$$h(z; c) = b_1^{(2)} + \sum_{j=1}^{4} \psi_{1j}^{(2)}\left(z_j^{(1)}\right) = (T_2 \circ T_1)(z).$$

ψ: learned scalar curves; b: node biases; T_1, T_2: first and second layers.

How each scalar edge is represented

An affine trend plus localized Gaussian basis functions

$$\psi(x) = \lambda x + \sum_{j=1}^{J} \xi_j G_j(x), \qquad G_j(x) = \exp\left[-\frac{(x-\mu_j)^2}{2\rho^2}\right].$$

$\lambda \in \mathbb{R}$: learned slope; $\xi_j \in \mathbb{R}$: learned basis weights.

Fixed Gaussian centers and bandwidth

J: bases per edge; μ_j: location of Gaussian j; ρ: its width.

$$\mu_j = -1.5 + \frac{3(j-1)}{J-1}, \quad j = 1, \ldots, J, \qquad \rho = \frac{3}{J-1}.$$

Centers are equally spaced on $[-1.5, 1.5]$ in each layer's input coordinate; the bandwidth equals their spacing.

Chosen architecture

Two $[1, 4, 1]$ KANs use transformed inputs to resolve the wage tail and $J = 33$ bases per edge, selected by validation from $\{9, 17, 33\}$ (554 parameters).

Training windows: one start and q targets

q counts future observations matched in one simulation

b: training window; $y_{b,i} \in \mathbb{R}^3$: observed (ω, e, d) at local time $i\Delta t$.

$$\underbrace{y_{b,0}, \ldots, y_{b,q}}_{q+1 \text{ observed states}}, \qquad H = q\Delta t, \qquad \Delta t = 0.1 \text{ years}.$$

The predicted path $\widehat{u}_b$ uses the current KAN coefficients.

Example: $q = 4$ gives five observations and $H = 0.4$ years

Time (years)	0.0	0.1	0.2	0.3	0.4
Observed state	$y_{b,0}$	$y_{b,1}$	$y_{b,2}$	$y_{b,3}$	$y_{b,4}$
Simulated state	$y_{b,0}$	$\widehat{u}_b(0.1)$	$\widehat{u}_b(0.2)$	$\widehat{u}_b(0.3)$	$\widehat{u}_b(0.4)$
Training role	Initialize	Compare	Compare	Compare	Compare

RK4 uses a 0.0125-year step: 8 solver steps per observation interval, hence 32 steps for $q = 4$.

State-only identification: the trajectory loss

A candidate vector field uses the two fitted responses

$$\widehat{F}(u; c) := F\Big(u; \widehat{\Phi}(\cdot; c), \widehat{\kappa}(\cdot; c)\Big).$$

Take a batch $\mathcal{B}$ of B windows: each has q observation intervals and $q+1$ recorded states.

$$\dot{\widehat{u}}_b(t; c) = \widehat{F}(\widehat{u}_b(t; c); c), \qquad \widehat{u}_b(0; c) = y_{b,0}, \qquad H = q\Delta t.$$

Minimized functional: normalized trajectory mismatch

$$\mathcal{L}_{\mathrm{traj}}(c; \mathcal{B}) = \frac{1}{H^2}\,\frac{1}{3Bq}\sum_{b=1}^{B}\sum_{i=1}^{q}\left\|S^{-1}(\widehat{u}_b(i\Delta t; c) - y_{b,i})\right\|_2^2.$$

Minimize its average over training windows with respect to c.

Training protocol: $B = 64$, $q = 1$ for 150 updates, then $q = 4$.

Algorithm: training the wage and investment KANs

Input: state observations $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{val}}$.
Adam updates the two KANs' coefficients c to minimize $\mathcal{L}_{\text{traj}}$.

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1 Initialize c from observed state increments.

2 **for** $k = 1, \ldots, 600$ **do**

3 Draw a batch $\mathcal{B}_k$ of 64 training windows using the schedule on slide 10.

4 Simulate each window by RK4 using the current $\widehat{\Phi}$ and $\widehat{\kappa}$.

5 $c \leftarrow \text{AdamStep}(c, \nabla_c \mathcal{L}_{\text{traj}}(c; \mathcal{B}_k))$.

6 **end for**

Keep c_{best} with the lowest validation trajectory loss.

Output: the learned responses $\widehat{\Phi}(\cdot; c_{\text{best}})$ and $\widehat{\kappa}(\cdot; c_{\text{best}})$.

Experiment: Checking the KAN representation by direct fitting

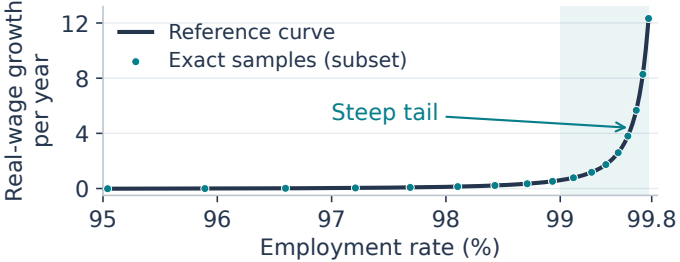
Fit exact response samples to test whether the KANs can reproduce the reference shapes.

Reference choices for the responses

Wage growth: $\dot{w}/w = \Phi(e)$. Investment/output: $I/Y = \kappa(\pi)$.

$$\Phi_*(e) = \frac{\phi_1}{(1-e)^2} - \phi_0 = \frac{0.04^3}{(1-0.04^2)(1-e)^2} - \frac{0.04}{1-0.04^2},$$

$$\kappa_*(\pi) = \kappa_0 + \exp(\kappa_1 + \kappa_2\pi) = -0.0065 + \exp(-5 + 20\pi).$$



Can the KAN reproduce this sharp rise?

Fit $\widehat{\Phi}(e; c)$ to exact pairs $(e_i, \Phi_*(e_i))$ across this tail.

Experiment: Objective and basis selection for direct fitting

Training data: sample the known response curves

To test curve recovery, use $n = 513$ exact pairs per function: $(e_i, \Phi_*(e_i))$ for wages and $(\pi_i, \kappa_*(\pi_i))$ for investment.

Objective: match the predicted and reference responses

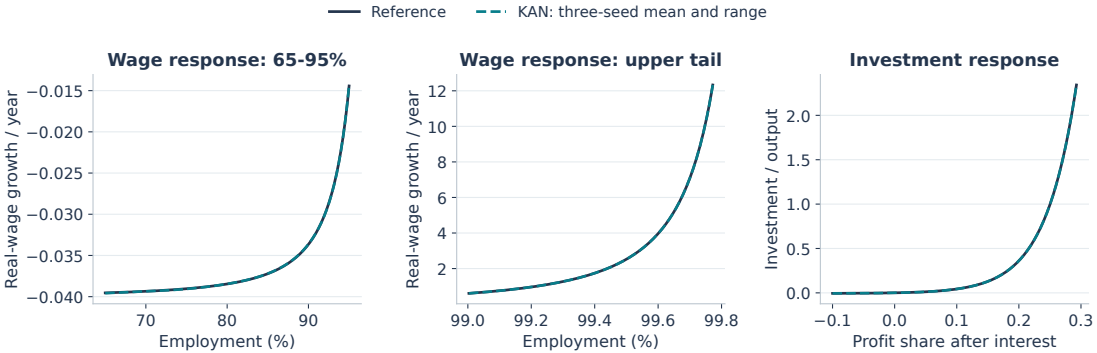
$$\mathcal{L}_{\text{func}}(c) = \frac{1}{2n} \sum_{i=1}^{n} \left[\left(\frac{\widehat{\Phi}(e_i; c) - \Phi_*(e_i)}{\sigma_\Phi} \right)^2 + \left(\frac{\widehat{\kappa}(\pi_i; c) - \kappa_*(\pi_i)}{\sigma_\kappa} \right)^2 \right].$$

$\sigma_\Phi, \sigma_\kappa$: reference training-value standard deviations, used to scale each response error.

Fit and validation

Fit without ODE solves; validate on 512 unused midpoints per curve using the same loss and scales. Select $J \in \{9, 17, 33\}$ by mean validation loss: $J = 33$.

Experiment: Direct fitting confirms capacity on the reference interval



Independent 64-path audit	Wage response	Investment response
Mean NRMSE	0.016%	0.006%

NRMSE = RMSE / SD(reference).

Return to state-only recovery and new initial states

Fit two sets of models using only state observations

Clean observations ($\eta = 0$) and noisy observations ($\eta = 0.01$); three seeds each.

Apply the algorithm on slide 11 to minimize $\mathcal{L}_{\mathrm{traj}}$; select the checkpoint with the lowest validation loss.

Freeze the recovered law and restart from unseen states

$$\dot{\widehat{u}}_m = \widehat{F}(\widehat{u}_m; c), \qquad \widehat{u}_m(0) = u^*_m(0) = u_{0,m}, \qquad m = 1, \dots, 16.$$

Evaluate recovery over five years

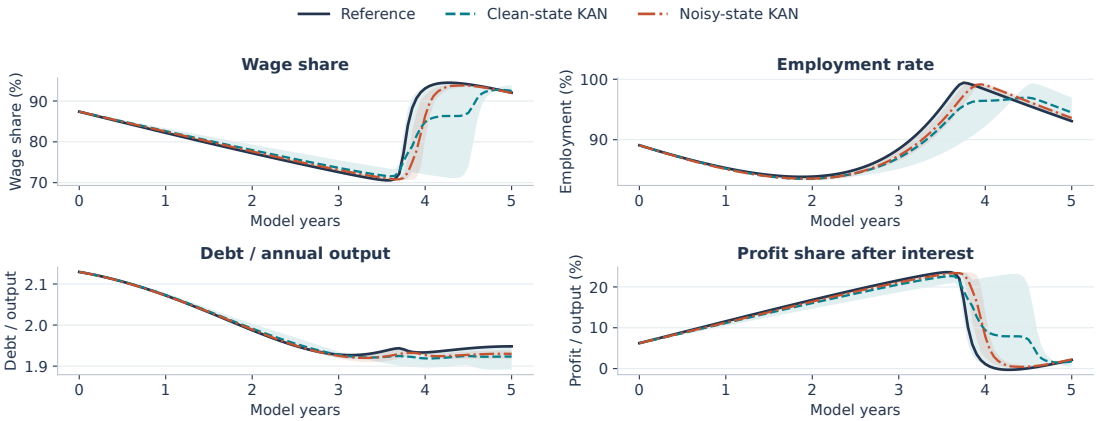
Compare predicted and reference wage share, employment rate and debt/output at $t_i = 0.05i$, $i = 0, \dots, 100$ (five model years).

$$\mathcal{E}_{\mathrm{test},m} = \left[\frac{1}{3 \cdot 101} \sum_{i=0}^{100} \left\| S^{-1}\big(\widehat{u}_m(t_i) - u^*_m(t_i)\big) \right\|_2^2 \right]^{1/2}.$$

$$\mathcal{E}_{\mathrm{test}} = \left(\tfrac{1}{16} \textstyle\sum_{m=1}^{16} \mathcal{E}^2_{\mathrm{test},m} \right)^{1/2}.$$

Recovered dynamics from a high-debt starting state

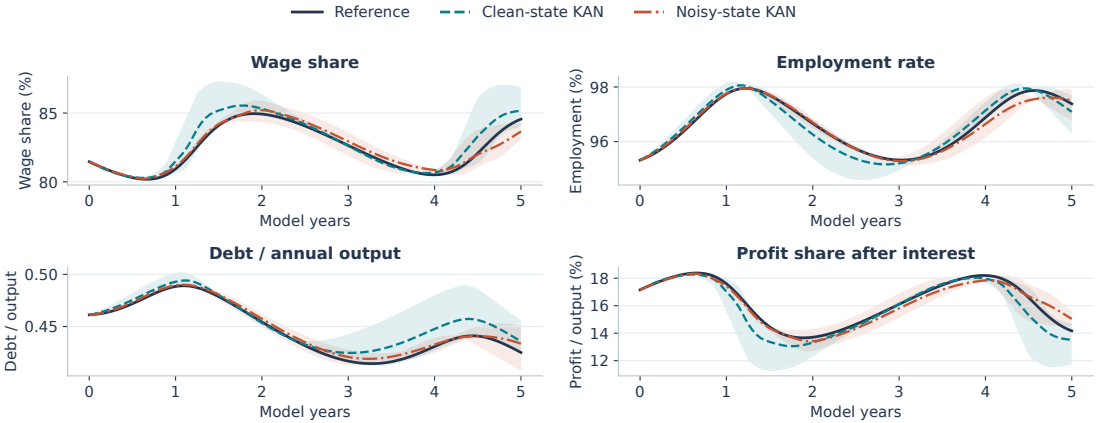
Example 1: $u_0 = (\omega_0, e_0, d_0) \approx (0.87, 0.89, 2.13)$.
KAN lines: three-seed means; bands: min–max.



Mean $\mathcal{E}_{\text{test},m}$ over 3 runs: Clean-state KAN: **0.238** Noisy-state KAN: **0.158**

Recovered dynamics around a smaller economic cycle

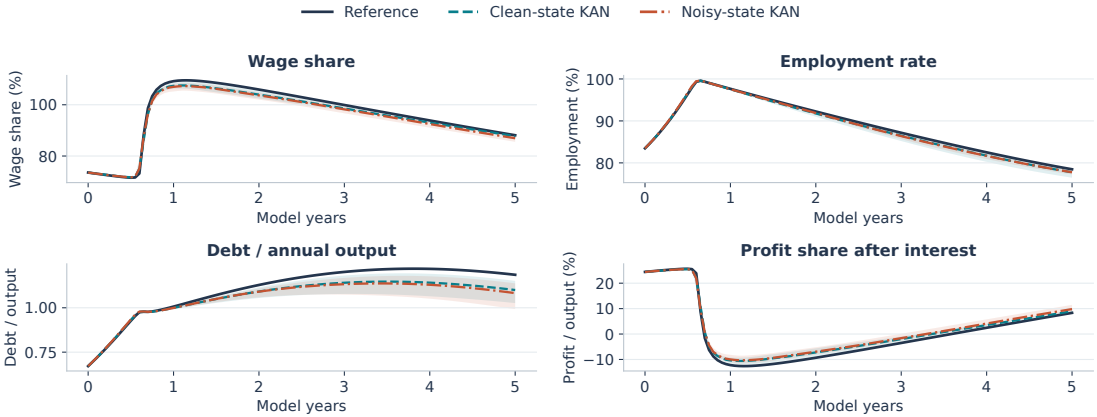
Example 2: $u_0 = (\omega_0, e_0, d_0) \approx (0.81, 0.95, 0.46)$.
KAN lines: three-seed means; bands: min–max.



Mean $\mathcal{E}_{\text{test},m}$ over 3 runs: Clean-state KAN: **0.047** Noisy-state KAN: **0.044**

Recovered dynamics from lower initial employment

Example 3: $u_0 = (\omega_0, e_0, d_0) \approx (0.74, 0.83, 0.67)$.
KAN lines: three-seed means; bands: min–max.



Mean $\mathcal{E}_{\text{test},m}$ over 3 runs: Clean-state KAN: **0.113** Noisy-state KAN: **0.121**

Conclusions

Inverse problem: recover the unknown responses

KANs reconstructed the wage and investment responses from observed state trajectories in the synthetic economy.

Forward Cauchy problems: simulate economic development

Each learned law was used without retraining to simulate wage share, employment, debt and profit from 16 unseen initial states over five years.

A tool for recovering dynamics

KANs can be a useful tool for recovering unknown dynamical relationships from observed trajectories in different contexts.

References

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